



TAKNEEK OnSpot Problem Statement



100 Points

Project: Tight-Fit

Time Limit: 3 Hours **Team Size:** 4 Members(2Y25s and 2Y26) **Duration:** 3hrs

The Scenario

The Robotics Club IIT Kanpur is gearing up for the International Rover Challenge (**IRC**). The suspension is built, the chassis is welded, and the autonomous codebase is mostly compiling without fatal errors. But there is a major issue: the rover's electronics currently look like a spaghetti monster exploded on a breadboard. Loose jumper wires are disconnecting every time the rover hits a bump.

The Club Secretaries have no idea how to make a schematic and route a PCB. To save them from eternal embarrassment (and to prevent the rover from catching fire on day one), your 4person engineering strike team has been drafted. You have exactly 3 hours to turn this rat's nest into a single, beautiful, and robust main control PCB. You are free to choose how many layer of PCB you want to design.

You are free to use the **internet and LLMS** but **online AI PCB generators are strictly prohibited**. The communication about the PS **must not** be shared with anyone outside the event area through any mode nor any help from anyone outside the solving area can be taken. Any team if found using AI PCB generators/ will be awarded with $S_{\text{Total}} = -100$.

The Arsenal (Bill of Materials)

You must design a custom PCB to house the following components. You decide the resistors, capacitors, and terminal blocks needed to bring it all together.

- 1x ESP32 DevKit V1
- 1x L298N

- 1x NEO-6M GPS Module (UART). *State on your documentation whether your specific breakout's TX/RX logic level is 3.3V or 5V — check the module's onboard regulator/level.*
- 1x HC-SR04 Ultrasonic Sensor
- 1x 4-Channel Bi-Directional Logic Level Converter Module (BSS138 based)
- 1x LM2596 or MP1584 (you can choose only one among them)
- 1x XT60 Connector for an 11.1V (3S) LiPo Battery
- 1x SPST Toggle/Rocker Switch (rated for at least **3A**)
- 1x **2A** Automotive Blade Fuse and PCB-mount Fuse Holder

Important Note: *The ESP32, GPS, Ultrasonic Sensor, Buck Converter, Logic Shifter, and Motor Driver are pre-built modules. You must provide suitable header pins, JST connectors, or screw terminals on your PCB so these modules can be mounted or wired cleanly to the main PCB. The ESP32 must sit completely on the PCB with the help of header pins. You can assume all types of resistors and capacitors are available in infinite quantities.*

The "Rules of the Garage" (Design Constraints)

1. The Power Distribution Architecture

Your PCB must clearly define, route, and label four distinct power domains. Mixing these up will fry the rover.

Protection hierarchy rule: every stage in the power path must be rated to *survive* the current required to trip the stage before it. From upstream to downstream, ratings must satisfy: Fuse rating (2A) $\leq$ Switch rating (3A) $\leq$ Trace ampacity (2A continuous). The fuse is always the weakest link and must blow before the switch or traces are damaged.

- **The High Voltage Line (VBAT / 11.1V – 12.6V):** The positive terminal of the XT60 connector must route directly to the 2A fuse, then through the Main Power Switch. Only after the switch can this switched 11.1V line be distributed to the Motor Power input (12V) of the L298N, the Buck Converter input, and the battery telemetry voltage divider.
- **The 5V Line:** Originates from the Buck Converter output. This must power the ESP32 (via its VIN/5V pin), the NEO-6M GPS, the HC-SR04 Ultrasonic Sensor, and the High-Voltage reference pin (HV) of the Logic Level Converter.
- **The 3.3V Line:** Originates from the ESP32's onboard regulator (via the 3V3 pin). This must power the Low-Voltage reference pin (LV) of the Logic Level Converter.
- **The Common Ground (GND):** All components must share a common ground. You must implement a continuous Ground Pour (Copper Pour) on the bottom layer.

2. Form Factor & Physical Rules

- **The maximum** size of the PCB is 80mm x 60mm.
- It **must** have four 3mm mounting holes in the extreme corners (inset by 3mm from the edges).
- All passive components (resistors, capacitors) you add to the board must be Surface Mount Devices (SMD) in 0805 or 1206 packages. This is a hard requirement — a board using any through-hole resistor or capacitor (excluding connectors, switches, fuses, and module headers) will be rejected outright and receive zero marks for the Engineering Reality section, not partial credit.

3. High Current Reality & Live Telemetry

- **Power Traces:** The XT60, Fuse, Switch, 11.1V, and 5V power lines must be routed with a track width capable of safely carrying **2 Amps continuous current**. Justify your trace width using IPC-2221 (or equivalent) in your math sheet.
 - **Battery Telemetry:** Route a voltage divider from the switched 11.1V battery line to an analog pin on the ESP32 to monitor battery life.
 - **Constraint A (Safety):** At $V_{in} = 12.6V$ (fully charged), V_{out} must not exceed 3.0V — i.e., you must design in at least $\sim 0.3V$ of margin below the ESP32's absolute 3.3V ADC limit, to account for battery overcharge tolerance, resistor tolerance, and supply noise.
 - **Constraint B (Usability):** At $V_{in} = 12.6V$, V_{out} must be no lower than 2.5V. This forces you to use the ADC's usable range efficiently rather than gaming the safety check with an overly conservative divider that wastes resolution.
 - Show both bounds explicitly in your math sheet.
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Your Deliverables

Submit all files in a single ZIP folder through the google form that will be floated.

1. **Schematic:** A clean, logical schematic that passes ERC. The four power domains must be clearly visually separated or labeled. All signal connections (L298N control pins, GPS UART, ultrasonic TRIG/ECHO) must be traceable end-to-end.
 2. **PCB Layout:** A fully routed PCB Layout that passes Design Rules Check (DRC) with zero errors.
 3. **The Math Sheet:** A single PDF or text file showing calculations for: (A) the 12.6V-to-3.3V battery telemetry divider (showing both the safety upper bound and usability lower bound), and (B) the 2 Amp power trace width justification (IPC-2221 reference or equivalent).
 4. **Pinout & Documentation:** A brief manual for the software and mechanical teams detailing every pin connection (power *and* signal), the board layout, level-shifter channel allocation, and the buck converter tuning warning.
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Point Distribution & Mathematical Scoring (Total = 100)

Your final score is the sum of four discrete grading functions:

$$S_{Total} = S_{Sch} + S_{PCB} + S_{Eng} + S_{Doc}$$

1. Schematic Score (S_{Sch}) — Max 30 Points

$$S_{Sch} = (8 * B_{pwr}) + (8 * B_{logic}) + (7 * B_{signal}) + (3 * B_{safe}) + \max(0, 4 - 2 * E_{ERC})$$

- $B_{pwr} \in \{0,1\}$: 1 if all 4 power domains are isolated and connected to the correct pins, 0 if any crossover occurs.
- $B_{logic} \in \{0,1\}$: 1 if all required 5V $\leftrightarrow$ 3.3V signals (HC-SR04 ECHO, and GPS UART if 5V-logic) are correctly routed through the Logic Shifter to the ESP32, else 0.
- $B_{signal} \in \{0,1\}$: 1 if every L298N control pin (IN1–IN4, ENA, ENB) and both GPS UART lines are connected to distinct, valid ESP32 GPIOs, else 0.
- $B_{safe} \in \{0,1\}$: 1 if the safety sequence (XT60 $\rightarrow$ Fuse $\rightarrow$ Switch $\rightarrow$ System) is strictly followed with the fuse rated $\leq$ switch rated $\leq$ trace ampacity, else 0.

- E_{ERC} : Total number of Electrical Rules Check errors. **Warnings for unrouted/floating nets, missing net-class assignments, or floating power-flag pins are counted as errors for scoring purposes** — only purely cosmetic/style warnings are ignored.

2. PCB Layout Score (S_{PCB}) — Max 40 Points

$$S_{PCB} = 15 * (N_{routed}/N_{total}) + \max(0, 10 - 2 * E_{DRC}) + (5 * B_{size}) + (5 * B_{holes}) + (5 * B_{pour})$$

- N_{routed}/N_{total} : Ratio of successfully routed nets to total required nets. N_{total} is defined as the count of distinct electrical nets in the final schematic after ERC, i.e., one net per galvanically distinct node — all GND pins collapse into a single GND net. This count must be stated by the team and is verifiable against their own schematic netlist.
- E_{DRC} : Total number of Design Rules Check errors (e.g., clearance violations, overlapping footprints, unrouted nets shown as ratsnest).
- $B_{size} \in \{0,1\}$: 1 if PCB dimensions are $\leq 80\text{mm} \times 60\text{mm}$, else 0.
- $B_{holes} \in \{0,1\}$: 1 if exactly four 3mm holes exist in the corners (inset 3mm), else 0.
- $B_{pour} \in \{0,1\}$: 1 if a continuous GND copper pour exists on the bottom layer, else 0.

3. Engineering Reality Score (S_{Eng}) — Max 20 Points

$$S_{Eng} = (10 * B_{trace}) + B_{THT} * (5 + \sum_{i=1}^5 L_i)$$

- $B_{trace} \in \{0,1\}$: 1 if all power traces (1.1V, 5V) are routed at a width $\geq$ the mathematically required width for 2 Amps, else 0.
- $B_{THT} \in \{0,1\}$: 1 if zero through-hole passives (resistors/capacitors) are used, **0 (and this entire bracket is zeroed) if any are used**.
- $L_i \in \{0,1\}$: 1 point for each required silkscreen label present: (1) Battery +/-, (2) 3.3V, (3) 5V, (4) VBAT Test Point, (5) Switch ON/OFF.

4. Documentation & Math Score (S_{Doc}) — Max 10 Points

$$S_{Doc} = (3 * B_{div}) + (3 * B_{calc}) + (2 * B_{pinout}) + (2 * B_{sigmap})$$

- $B_{div} \in \{0,1\}$: 1 if the voltage divider calculation proves $2.5V \leq V_{out} \leq 3.0V$ when $V_{in} = 12.6V$, else 0.
- $B_{calc} \in \{0,1\}$: 1 if the trace width calculation (or standard IPC-2221 table reference) for 2A is explicitly documented, else 0.
- $B_{pinout} \in \{0,1\}$: 1 if the power-pin pinout guide and buck converter warning are present and readable, else 0.
- $B_{sigmap} \in \{0,1\}$: 1 if every signal connection (L298N control pins, GPS UART, ultrasonic TRIG/ECHO, and level-shifter channel allocation) is mapped out with specific GPIO numbers, else 0.

End of Problem Statement

All the Best

For any queries, the PS lead can ask any of the coordinators present during the event. The clarification will be provided to all pools together.

